Research Paper: Reading Indus-Harappan Script with tools of a new paradigm

Title: Indus-Harappan Script: A Multi-Disciplinary Analysis of Linguistic Continuity and Urban Topology

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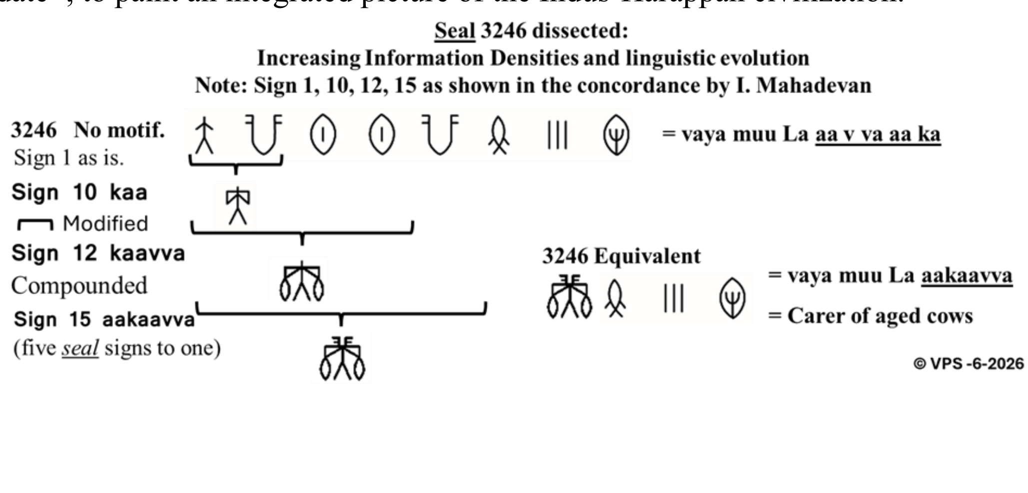
1. State of Research

The hundred years research into the Indus-Harappan script has resulted in it being termed an Enigma and Conundrum, and others claiming decoding it, or it being not even a language. Bringing pre-history into verifiable history is a slow, methodical, and evidence-based research, which effort can be repeated by others of even-tempered minds, to verify, clarify, modify, even oppose, and bring about new discoveries – a welcome change by a new paradigm, in this research field.

The new paradigm needs to provide a space for those even-tempered researchers to evaluate and recreate this research. A paradigm shift is detailed in the "Indus Script Research Keys" by the author of **"Indus Script is a Language"**[i], and other books. **Spoken language long predates writing**, **so the task is a "sound translation"** using a living spoken form, not a hunt for a Rosetta Stone. By prioritizing the reconstruction of the spoken language over visual *'decoding'* of signs in an unbound space, new research suggests testing a profound continuity between Harappan language and script, and the earliest **spoken form** of a classical language. This report provides a full review of this methodology, starting with the Tamizi script and its apparent similarities to some Indus script, and complements with known facts, and evaluates its implications for Harappan urban life. An even minded seeker can repeat the research. Newer pathways are proposed for completing this monumental work in Southeast Asian Archeology and Linguistics.

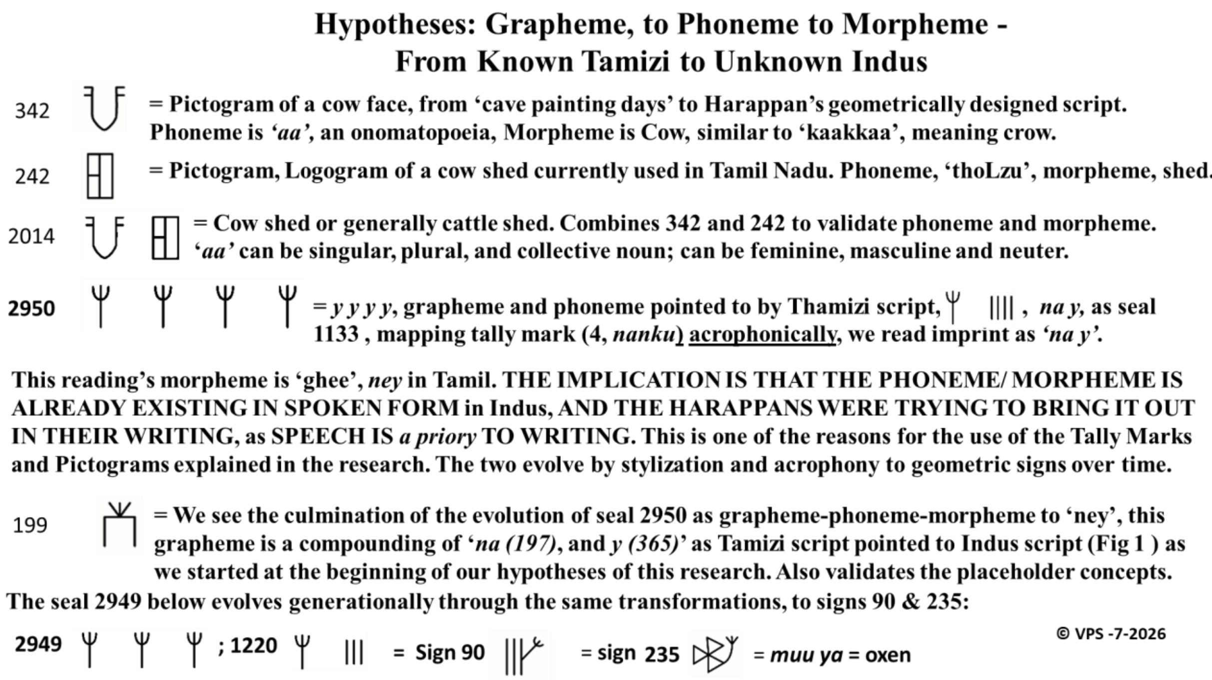
2. <u>The "Cow Face" Hypothesis</u>[ii] : Mr. Iravatham Mahadevan's call to identify 'jar' sign.

Most prolific sign in the Indus-Harappan script known as 'JAR' is the 'once upon a time pictogram' in cave paintings of a cow – a 'U' with two prominent serifs on open end of each column, depicting what appears as a cow's face, two horns, and two ears, a geometric grapheme (sign 342). The phoneme is, '***aa',*** reminiscent of the animal's sound of 'maa..', seen in ancient and modern Tamil as 'aa' (ஆ), an onomatopoeia, meaning its morpheme, cow. This is similar to the name 'kaakkaa', the common crow. The ancient Tamil grammar treatise[iii] establishes that the sound form is ancient, long before the written script came about, a view supported by Historic Linguists. By establishing this form, grapheme, phoneme, and morpheme, as suggested by Mr. Iravatham Mahadevan, this research reads seals related to animal husbandry, using Tamil phonemes in Tamizi discovered to date[iv], to paint an integrated picture of the Indus-Harappan civilization.



In this slide we see how signs progressively evolve in managed language script. A meaningful syllable is read the same in a seal and imprint, though a seal is scribed from right to left and its imprint read 'correctly' from left to right, as the language. Here concordance sign 10 uses the 'aa' (7^{th} of the **seal**) as a modifier, which integrates with sign number 1, ***ka***, becoming '***kaa'***, sign 10. It takes two additional graphemes, '***v***' and '***va*',** [pointed to by Tamizi script] to become sign 12. Sign 12 takes on the next '**aa**', 4^{th} in **seal**, morpheme being, cow. Sign 15 thus integrates last five graphemes of 8 signs of **seal** 3246, increasing information density and efficiency in scribing, a written language in evolution. Other examples of ***Schematization or Stylization, and Acrophony*** are shown in figures 4,5.
The jar is the most frequent sign in the Indus Script and hence the discovery of its value must be the first step - ***Harappa.com; Iravatham Mahadevan.*** We establish the baseline as shown in Fig.1 at the end to show the appearance of link between Tamizi and Indus scripts to test. Once we have this starting point, we add other signs meaningfully to build up our Indus vocabulary.

3.0 Key research hypotheses encapsulated below: Need to understand before proceeding



4.0 Mapping the sounds and Comparative study of inscriptions: Fig. 1, 2, 3

Our search for a Rosetta Stone indicates a need for a boundary within which we want to read the script, otherwise we are working in an unbound space. This reconstruction effort identifies 10 vowels and 18 consonants of the Indus script which is suggestive of the *Tamizi* script, that came to use in subsequent millennia. The character evolution shown by the study establishes a process of geometric abstraction, where earlier work-around pictograms are simplified, to meet the demands for standardization for efficient writing. The process of ***Schematization or Stylization and Acrophony,*** are demonstrated by the new tools (Fig 4,5). The consistency between Indus signs and the Tamizi script suggests a resilient cultural underpinning, despite millennia of time distance. Of the 30 graphemes of Tamizi, more than 14 of them are either ***a match or with slight variations*** due to change in medium and growth factors. This we can see in the loanwords in the first Indo-Aryan writings, supporting a persisting linguistic substrate in the Indus-Harappan region, predating the Indo-Aryan first ***writings***. The research demonstrates the standardization of Harappan writing visible in the uniformity of signs across 1.3 million square kilometer area for nearly 2000 years, in dozens of scribing centers across many generations, parallelling the broader affinity for standardization we see. This allows the testing of the idea of a sole language in the scripts.

5.0 <u>Geometric Evolution of Script</u>

The Harappans used pictograms, single and compounding letters, tally marks, and sound modifiers to increase ***information densities***, as demonstrated by this research. The use of iconography - motifs - in seals are tested for magnifying the message yet to another level. Also, this research gives the insight that the script was in active evolution by stylization and acrophony over generations. During this time, the pictograms, including tally marks, were converging to efficient geometric script forms. The driving force for this evolution was the sacrificing of clarity of pictogram to standardization of geometric form, for efficient scribing, on durable media like steatite stone chips, copper, clay, and pottery.

6.0 <u>Acrophony - The Tally Mark as Phonetic Substitution</u>: Fig 4,5,6

One of the most innovative methodologies proposed, is the reinterpretation of the tally marks to bring out the information density of the Indus script. The traditional scholarship of counting and accounting besides, the tally was a phonetic place holder or a 'work-around', to bring out sounds that yet lacked geometric signs, particularly short and long form of Tamil vowel, 'u' and its syllables (உ ,ஊ), missing in the corpora. ***Some*** of the pictograms are tested as functioning in the same mode. Name of numbers in Tamil begin with sounds (Acrophonical) that are homophonous and embed grammatical markers. By tally mark four (***nanku***, ***na***) a scribe could phonetically encode a word for ghee (nay or ney). We see signs 95, 104, preceding sign 162 for 'Y', sounding '***nay*** or ***ney'***, meaning ghee. This acrophony allows for a flexible and denser communicative writing. It also addresses the observation, that sign repetition within individual inscriptions is relatively low; this density enabled by modifiers and work-arounds, eliminates repetition, simplifying syllabic communication, the hallmark of a developed written language. One letter 'words' (>40 of them) to four letter words abound in Tamil and can enable poems with single consonant with all its syllables as literature demonstrates.

7.0 <u>Meaning of Iconography in Motifs (Topology of Indus cities)</u>

One of the efforts in research of the Indus-Harappan script, is the interpretation of the iconography or motif in seals. Motif-as-street-name is a novel hypothesis, motifs being furthest from the script itself. This research into urban topology mapping is this study's attempt to anchor the text to physical archaeological contexts. The research proposes a **"deliberate, highly ordered, urban topology hypothesis"**, suggesting and trying to verify the functional role of **some** of the motifs, as identifiers of the street and sections of the city structure. This would have allowed residents to navigate the city and understand the origin, destination, availability of goods, and applicable laws or ordinances specific to products, services, and places, and facilitate basic interpersonal connections as modern topology does. However, some motifs show no connection with the inscriptions found on seals that need added explanations. Also, we see motifs waning in the declining cities phase of this civilization, that need be evaluated within the hypothesis.

7.1 <u>The Unicorn and the Market Common</u> [v]: Table - 2

The unicorn motif appears in about 60% of seals ***with*** iconography. This research proposes unicorn as the emblem of a **'Market Common',** a commercial square. Seals with this motif show available goods, dairy products, cattle, special facilities, and specialized services. Products like fresh milk and milk products, emergency milk, variously described cattle, and home for aged cattle as service. Seals[vi] with the unicorn motif feature inscriptions related to "just expressed fresh milk" (seals 3023, 2358, *didiir aa*), "very good ghee" (seal 1076, *thava ne-y*), and "skilled cow carers and field guards" (seal 2082). The prevalence of the unicorn suggests that the Market Common was the administrative and economic heart of the Harappan cities, a space where standardized weights and measures could be widely employed, and where public laws and governing values could be prominently displayed.

7.2 <u>Street-Level Identification</u> : Table - 2

Study proposes that motifs were used as 'biodiversity street names', as in many urban contexts.
Elephant Street: Location of ghee shed (seal 2648) and producer of sesame 'ghee' (seal 2127), (eNNey in Tamil). Bull Street: Shows market activities including milk, housing for oxen, displayed banners, etc. Tiger Street: Associated with high-value cattle shed (seal 1386), a composite motif combining many streets or many sheds in a given street. Gharial: This seal 2864 links riverine activities or laws governing the usage of the river facilities.
This hypothesis explains the existence of **identical inscriptions with different motifs**. Seals 2444 and 2864, both carry the same command '*aNNa(l) kai aaNai'* ("Order of the Leader"), but one features a unicorn for Market Common jurisdiction, and the other a Gharial jurisdiction, under the urban topological hypothesis, these appear as the same laws applicable to two locations.

7.3 <u>The Universal Application of Motif-less Seals:</u> Table - 2

Some seals without motif can be interpreted as universal, like laws or ordinances, that apply to the whole domain, like the city. Citywide curfew (seal 4440), law from a farmers' guild (seal 4371), or a skill certification for a worker (seal 4284), are not location restricted.
Ethical foundation: Some ***read*** seals shed light on political ideology and social value of this civilization. Unlike Egyptian and Mesopotamian models of divine kingship and forced labor, the Harappan society shows self-directed ethics, collective welfare, and minimal or no conflict.
As an example, seal 2234 shows a bull motif in Iravatham Mahadevan's concordance (as in RMRL research tool), **"The world is a Big Home".** The study interprets this as displayed at the Bull Street, as a guiding banner of value of ***Harappan Governance***. This means that the 'guards' mentioned in many a seal means only one thing: CARE - caring for the city, the people, cattle, cultivated field, and the neighbor. This explains lack of military imagery, weapons, and dictatorial leaders and their accouterments, but supports a concept of a peaceful, farming led ***literate democracy,*** guided by enlightened leaders and level headed advisors as read in seal 1110, **'respected equanimous great leader'**.

8.0 <u>The Safety Net: "Panic Milk" and Elder Care:</u> Table - 2

Most compelling evidence of social maturity of Harappans shows up in seal 4718, 'panic milk for children'. A seal without a motif, seen as an ordinance mandating fresh milk from "on-demand cows", (3016 ***didiir fresh milk***, 3023 ***didiir aa***). These seals (with few similar ones) have unicorn motif, implying that mandate applies to milk supply at the Market Commons, to meet the needs of infants and children at **all times**. The stacked schedule of milking the ***'instant cows',*** can bridge the availability of fresh milk, to the next general milking. We note a milk cow shed at the Market Common housing milk cows and carers, and seals declaring 'just expressed milk's availability. Seal 1425, 'illam vaya muu thani', **'Home for the aged and alone', s**eal 5119, 'muu da aa thoLzu', '**shed for aged cattle', and** seal 3246 'vayamuula aagaavva', **carer of aged cattle**, all indicate a mature, compassionate civilization that valued life beyond its economic utility, verifiable by motif on the seal. 'Shed for aged cattle', has unicorn motif and we see no motif on the 'home for the elders' seal! Aged cattle beyond productive years were sheltered to self-sustain and yield manure, a valuable fertilizer, showing a sophisticated integration of sustainability. The compassionate care of elders seal, without motif, shows it is not a business, but a ubiquitous welfare service. All of these scripted in handful of signs, in precise language long matured, to express these ethical ideals nearly 5,000 years ago.

9.0 Governance and the Dholavira Signboard: Fig 4, 5, 6

Harappans indicated their leader as aNNa or aNNal, currently in use in Tamil– meaning Elder and Leader. This provides a new interpretation of the famous Dholavira signboard reading as the **'residence of the king's advisors and grain store'**, paralleling the seals 1110, 2503, and 8117. Seal 1110 identifies a **'respected equanimous great leader'**, who is found at the Market Common, likely functioning as an adjudicating mediator in commercial disagreements. The other seals amplify the message of grain store by indicating the 'Care of the village' (seal 2503) and related laws (seal 8117) of the KoLzi village. Most of the seals ***read*** explain how a mature, literate, compassionate civilization lived in peace for over 700 years, in its mature phase.

10.0 Critical Analysis of Competing Hypotheses

The methodology presented in the "Research Keys", provides rebuttal to several dominant theories in Indus studies, most notably the "non-linguistic" thesis, championed by Steve Farmer (U Penn), Richard Sproat (U of Illinois), and Michael Witzel (Harvard U), well known in their field.[vii] The claims of 'non-linguistic' thesis including **lack of linguistic evolution over centuries and singletons,** is dealt with by the research.

10.1 Rebutting the Non-Linguistic Thesis

The Farmer-Sproat-Witzel thesis argues that the brevity of Indus inscriptions and the lack of complex syntax indicate that the signs were not a true script encoding speech, but rather political or religious emblems. They point to the high number of signs occurring only once - "singletons"- and lack of linguistic evolution over time as evidence against a language-based system. Argument is that the elites feared loss of control over the population if language is universal.

10.2 The "Research Keys" provides a robust rebuttal to these points: Fig 4, 5, 6

Brevity, linguistic evolution, and Professionalism: The brevity of the seals is a function of their purpose as identifiers and permits (as many as 9 differing uses), not a limitation of the script itself. Seals 2648, 2949, 2950, (2322), 2127, signs 90 & 235, all show how, (8), 4, 3 repeat signs of seals become 2 and 1-sign graphemes, creating information densities of singletons. Research demonstrates how tally marks and pictograms evolve by ***schematization and acrophony towards efficiency and information densities***. Dissected seal 3246 (fig 3) demonstrates other linguistic evolutions over time. The time distance for this evolution can be verified by stratigraphy data (or serialization, relative dating methods). Work by Rao and Yadav (on entropy) and Yadav et all (n-grams) are seen supporting these language arguments. Long-form writing ***likely*** existed on perishable materials recorded later around 300 BCE, which have not survived time, but definitely indicated by seals like 2617, 'buttermilk available HERE'. People are ***not*** reading the small permit seal, but a large banner or sign board declaring the product availability at the market.

Singletons: Many unique signs are symbols created by compounding and adding modifiers. In Tamizi and Tamil a single vowel using a single modifier can create all 18 syllables of consonants. We see more than 40 one letter words in Tamil, so a singleton may not be an issue. It is the difficulty in reading and finding their meaning in a syllabic language as demonstrated.

Dravidian, Indo-Aryan links: There is also hypothesis for a Proto-Sanskrit or Indo-Aryan origin for the script. Sanskrit, acclaimed oral language initially without script, cannot claim origin for the Indus-Harappan script, which predates them. However, the work of Iravatham Mahadevan and Asko Parpola is extended and enhanced by this research, by providing direct phonetic bridge to spoken ancient version of Tamil, which exists in the literature and seen in current usage, culture, and practices. By mapping signs, graphemes, to Tamil phonemes and morphemes, it provides a nearly consistent and reproduceable model, that accounts for both the administrative needs of the civilization and its underlying ethical philosophy, which has parallels in its ancient literature. Any researcher can approach this work by using the language for identical results, and even, improvements. Variations noted are explained.

11.0 Technical Nuances of Script Construction (fig 3)

The Indus scribe used several other sophisticated language mechanisms, to expand the phonetic range without increasing the number of basic signs, that remains around thirty.

11.1 The Mechanism of Modifiers

Modifiers possess name and form but no sound of their own, functioning grammatically to alter the sound of signs, and syllables, to which they are attached. This design is preserved in Tamizi and Tamil, where a modifier keeps the form (grapheme) of a consonant, ***constant*,** even as producing varied syllabic sounds. One modifier frequently used in Indus script adds vowel 'e' sound, and another adds one or more suffixes. We can see sign 59 (la, La), progressing to 65 (eLa) and 60 (eLam), both meaning 'young', and to 66 ('illam') 'home'. Similar readings are seen in sign 216 change to 219 and then to 220. Modifiers allow for the creation of complex terms within the constraints of grammar and of a limited seal surface. Modifiers have name and form but no sound, and are not in the concordance.

11.2 Compound Integration and Labor Efficiency

In seal 3246, (under 2.0, cow face analysis) we saw the compound integration of signs, for efficient writing and the evolution of sign 10, 12, and 15. Seal 3246 uses five signs to say 'aakaavva' (cow carer) and the scribe creates sign 10, 'kaa', sign 12 'kaavva', and finally sign 15 'aakaavva' as a single sign. This linguistic evolution from linear to integrated writing is a key indicator of a mature, professionally managed evolving script over time, verifiable with stratigraphy and equivalent falsifiable testing.

12.0 Ways to Improve the Research and Future Pathways

While the **"sound translation"** methodology offers a compelling model for *decipherment*, several avenues for empirical improvement are identified.

Integrating the stratigraphy data: Systematically correlating the stratigraphic data with sign forms will show the developing phases of the signs – from Ravi, Kot Diji phases, to mature Harappan phase. The later showing dominance of simplified geometric signs. This will provide the chronology of script development shown. With questionable stratigraphy data, falsifiable testing may include **Seriation, Relative dating**, and other tools available to dating.

Spectroscopic analysis: Gas chromatography-mass spectrometry, (GC-MS), will help identify presence of milk and sesame lipids in shards found in proximity to seals reading ghee, sesame oil, buttermilk etc., to verify the validity of these seal readings. Shards with graffiti mark 'ghee', for example, pottery graffiti **seal number 2929**, **in the ASI concordance**, is an ideal test pieces with good confidence level, one independently falsifiable prediction. Even a small pilot would move the work from interpretation toward evidence. Many other steatite **seals or metal sealings** with milk product names inscribed, may not yield such verification as a porous pottery shard.

Comparative study of "Trade Jargon": Close to 150 **signs** (36% unknown to 64% 'known') are still unidentified to help 'read' remaining seals and their phonemes-morphemes. Future research involving Tamil Family of languages (Malayalam, Kannada, Telugu, Tulu, Brahui and others) to identify various trade or skills jargons -**potter, weaver, metal worker, jewelry maker, carpenter, boat builder** etc., to expand the understanding.

Expanding current Corpus: The current research of 100 years is centered around the discoveries of 10-15% of the Harappan territory excavated. Expanding excavations may yield other inscriptions – perhaps longer ones.

13.0 The study inferences:

"Indus Scripts Research Keys" can lead to many nuanced interpretations:

Harappans were beginning to record the sound of their already well developed spoken language. They used placeholders to meet gaps in signs, to help generate sounds **in use**. **T**he script was dynamically evolving to reduce the gaps in written script, similar to the first human speech, which was gradually replacing gestures, Historic Linguists can agree. Testing shows that placeholders - tally marks and pictograms - were replaced over time by sign designs, by **Schematization** and **Acrophony**, verifiable with stratigraphy, seriation and relative dating – falsification testing. **W**idespread literacy is verifiable with graffiti marks and presence of banners at markets. We can infer that literacy was not restricted to 'elite' but used by artisans, farmers, laborers, merchants and administrators. **L**anguage was helping the administrators with efficiency, accuracy, and speed, equally. **T**opology in Navigation: The street and market name hypotheses provide a functional explanation of **some** of seal motifs, pointing visual urban management, and integrated linguistic urban planning. **E**thical Administration: The content of the seals, emphasizing "the world is a big home", welfare for children, care for the elderly, and care for aged cattle – all paint a picture of a civilization that achieved stability through collective responsibility rather than coercion. **L**inguistic Continuity: This approach of phonetic mapping to ancient Tamil family of languages offers consistent and compelling explanation for the script's reading, suggesting the Indus Valley was the cradle of the Tamil family linguistic and cultural traditions, as supported by Mr. Asko Parpola and Mr. Iravatham Mahadevan.

The interdisciplinary approach as a new paradigm proposed by 'Reading the Indus Script with new tools', demonstrated the combining of epigraphy, linguistics, urban archaeology, anthropology and urban topology, represents a viable pathway to complete this study and scientifically validate **reading** of the world's most enigmatic writing. This also points to new directions for AI and Machine Learning tools. This multidisciplinary approach will bring the Harappans from prehistory into history, improving the research from the many not falsifiable concepts that have taken residency in the vacuum of history, since the discovery of this ancient civilization over 100 years ago.

A summary of the supportable hypotheses:

1. The Indus-Harappan Script is a grammatical syllabic language
2. The Script is linguistic notations, markings, and commercial seals of ownerships, approval, accounting, laws, ordinances, permits, and identifications
3. The Indus Script and the language spoken are one
4. The script is not a code, cypher nor a cryptogram- it reads as a language
5. The script is founded in the Indus, however connected with rest of the subcontinent[viii]
6. The Indus language is ancient form of Tamil, recognizable by today's speakers
7. The Indus script and the language are not Aryan, nor Sanskrit, nor any unknown language

Notes on figures:

In the script we see 'la' & 'na' having three sounds each: la, La, Lza (ல, ள, ழ); n, na, Na (ன, ந, ண). The sound families lead to some confusion of letters as used by scribes, stretching the existing grammar to appear as freedom. However, this freedom is not total nor real. When we see high frequency signs are constrained and when we research a random section of a page of the Iravatham Mahadevan's concordance, **a blind or held-out test**, Fig. 7, this truth bears out.
Data sets:

Notes on motifs, iconography, field symbols:

Unicorn: Studied seals show relationship to the Market Common hypothesis of the study.
Other animals: Elephant, Bull, Gharial, Composite animals, and Fabulous animals are pointing to street names, base on research hypothesis of Urban Topology. Some motifs may point to packaging seals identifying commercial information. Some 'no motif' indicates order, law, ordinances etc. controlling and applying to a wide domain. Seal with two different motifs indicates, permit to post message in different streets and same law applying to two different jurisdictions – urban topology explains. In the declining cities phase of the civilization, we see motifs diminishing in the seals, which need to be identified and read to explain within the urban topology hypothesis.

The following two tables list all the signs and seals explored in this research paper.
Tables:

1. Signs, graphemes, phonemes, morphemes, in Indus, Tamil, and English. Studied seals below use these graphemes, phonemes, and morphemes
2. Seals with Indus **readings and English translations**. Motifs show following insights: The seals with unicorn motif show reasons to belong in the Market Common, that do not show conflict with one another. The urban topology hypothesis has explained the

studied seals readings and their relationship to each other and other seals read in general.

3. In attached figures 1-6, all signs and seals of paper are explained though some *may not* appear in the table 1 and 2.
4. These figures focus on the dynamic evolution of the Indus script over generational time distances by **Schematization or Stylization, Acrophony**, **and** how other language mechanisms help in increasing information densities in this newly discovered **technology of writing the spoken language,** of the Harappans as it **evolved** in use.

The number of **seals** shown read by this research is over 200, but can exceeds 700 as they are representing 1, 5, 20, 40, 60, 130, and even 400[ix] seals each or samples, of same subject morpheme, and the claim can be higher. Some seal counts: Cow-300, milk-44, ghee- 24, ordinance- 128, children's food – 160, manure – 60. However, some graphemes are not assigned with phonemes and morphemes and not verified in seals ***readable***. We can 'read' with possible sounds of more seals but without grasping their full meaning, without further research on technical jargons of the time as this research so far demonstrates.

Table 1: Indus-Harappan Signs in this paper

SIGN #	Indus Grapheme	Tamil Phoneme & English sound	Morpheme & English Meaning	SIGN #	Indus Grapheme	Tamil Phoneme & English sound	Morpheme & English Meaning
1		க; ka		169		னி; ni	
8		கை,கய்; kai	கைத்து; hand	171		ண; Na	
10		கா; kaa		176		ண,ணை; Na,Nai	
12		காவ்வ; kavva	காப்போன்; carer	182		பண்ணாட; paNNatta	உழவர்; Farmer
15		ஆகாவ்வ; aakaavva	ஆ காப்போன்; cow carer	199		நெய்; nay	நெய்; Ghee
53		ம; ma		204		கோயில், koil	கோயில்; Temple
59		ல, ள; la, La		216		ச, sa, cha	
60		லம், லன், லர்;ளம்,ளன்,ளல், ளர்;lam,lan,lal,lar		219		சி; se	Syllable
65		இள; iLa	இளம்; young	220		சிஅம்,சிஅன்; Sivan,Sivam	
66		இல்லம்; illam	இல்லம்; Home	242		தொழு; thoLzu	தொழுவம்; cowshed
67		லா; laa		245		இட(ம்); ida(m)	Place; mark row & column
76		கோழி; KoLzi	கோழி ஊர்; KoLzi village	256		ண், Na	
86		ஒரு,ஓர்,ர்,ஒ,ஓ ;oru,oor,r,o,o o	One; also phonemes	261		ம், ம; m, ma	
87		இரு,ஈர்;iru	wo; also phoneme	290		டி; de	Syllable
89		மூன்று; மு,மூ mu,muu	Three; also phonemes	292		திடீர்; thideer	திடீர், தடால்; sudden
90		மூய; muuya	எருது; oxen	293		ட் ; it	letter
91		முரா; mura	மோர்; buttermilk	294		ட்ட, itta	Syllable
95		நான்கு,ந, நா; nanku,na,naa	Four; also phonemes	325		வ், வ; va	
125		த; tha		327		அரசனின்; arasana	அரசனுடைய; King's
137		த; tha		336		வ்ய; vya	Syllable
162		ய்,ய; y,ya		342		ஆ; aa	பசு; cow
165		ண் ய், NNai	எண்ணை; sesamum oil	343		அய்; ai	ஐ, vowel
166		ய் ண்; NNai	எண்ணை; sesamum oil	387		வய; vaya	வலிய; strong
167		மீ; mee	சிறந்த; good	391		தவ; thava	மிக்க; good
169		னி; ni		400		வ்வ' vva	Syllable

Table 2: Studied seals: Listed in seal number order as shown in the Concordance.

Seals / signs	Graphemes	Reading	Meaning	Acrophony in reading	Motif
1045		vaya(l) gaavvan	Carer of the field	n' as gender suffix - male	Unicorn
1076		thava thava ney	Very good ghee	Tally modifies by acrophony-to 'na'	Unicorn
1110		thava sa ma thaLa thava NNal	Respected equanimous great leader	aNNa','aNNal' - in current use	Unicorn
1133		nay	ghee	Tally becomes 'na' -acrophony	Unicorn
1220		muuya	oxen	Here tally acrophonically becomes 'muu' letter sign	Unicorn
1228		mi mi KoLzi aagaavva	v. good KolZi uur cow carer	Skill certification - no motif	Unicorn
1386		mi mi [thava] aa k thoLzu	v. good zebu cattle shed	Missing sign fittingly filled in w/o software	Compsote motif
1425		illam vay muu tha ni	home for the aged and alone	Tally stands for 'muuththa', aged	No motif
2082		Isa muu aagaavva	cow carer of 3 cows of Isa	Tally stands for count, 3	Unicorn
2127		eN y	oil (sesamum ghee)	Tally becomes 'eN' by acrophony	Elephant
2167		muuya thoLzu	Shed for oxen	Tally compounded with sign	Unicorn
2234		vay ya iir illam	world is a big home	Tally shortened - 'iir' (instead iru)	Bull
2322		nay aakaavva	ghee from cow carer	Sign count by acrophony – nay aagaavva	Bull
2358		didiir (dadaal) aa	instant cow (on demand)	Modifier increases density	Unicorn
2442		da tha ay vya mugar	Strong fivefaced deity	Pictogram brings 'u' sound - 'udaiya', a possessive	Unicorn
2444		NNal's aa Nai	leader's command	Same as 2864	Unicorn
2503		mi mi koLzi gaavva	v. good Kolzi uur guard	KoLzi is KoLzi Ur - refers to Mohenjo-daro	Unicorn
2617		ivva da t l vya mu r mura	buttermilk available here	Tally is mu or muu – acrophony	Unicorn
2648		mi mi na y thoLzu	v. good ghee shed	Tally is 'na' by acrophony	Elephant
2864		NNal kai aa Nai	leader's command	Same as 2444- different motif	Gharial
2929		nay	ghee	Tally reads as 'na' by acrophony	Pottery shard
2949		muuya	oxen	Count turns tally – then 'muu'	Circle symbol
2950		nay	ghee	Count turns tally – then 'na'	Circle symbol
3016		di diir ay laa iaLa vyamun	just expressed fresh milk	Tally to 'mu/muu' by acrophony	Unicorn
3023		di diir (da daal) aa	instant cow (on demand)	Increased info densities	Unicorn
3246		vaya muu La aagaavva	carer of aged cow	Tally reads – 'muu' for 'muuththa'- aged, acrophonically.	No motif
4284		koyil ida aagaavva	carer of temple cow	Pictogram brings 'u' sound - 'udaiya', a possessive	No motif
4371		malaaLa paNNada aa Nai	ordinace by farmers guild	No motif. Wide application	No motif
4440		iLa iru tta aa Nai	early night order	By acrophony, tally reads 'iru'	No motif
4718		ay ya ay o vyamun	panic milk for children	Tally to 'mu/muu' by acrophony. Law, no motif	No motif
5119		muu da aa thoLzu	shed for aged cattle	Tally to 'muu' for 'muuththa', 'aged' by acrophony	Unicorn
8117		KoLzi aa Nai	KoLzi uur ordinance	Village ordinance - no motif	No motif

Figures 1, 2, 3: Attempting validation of the research's language hypotheses:

Fig. 1

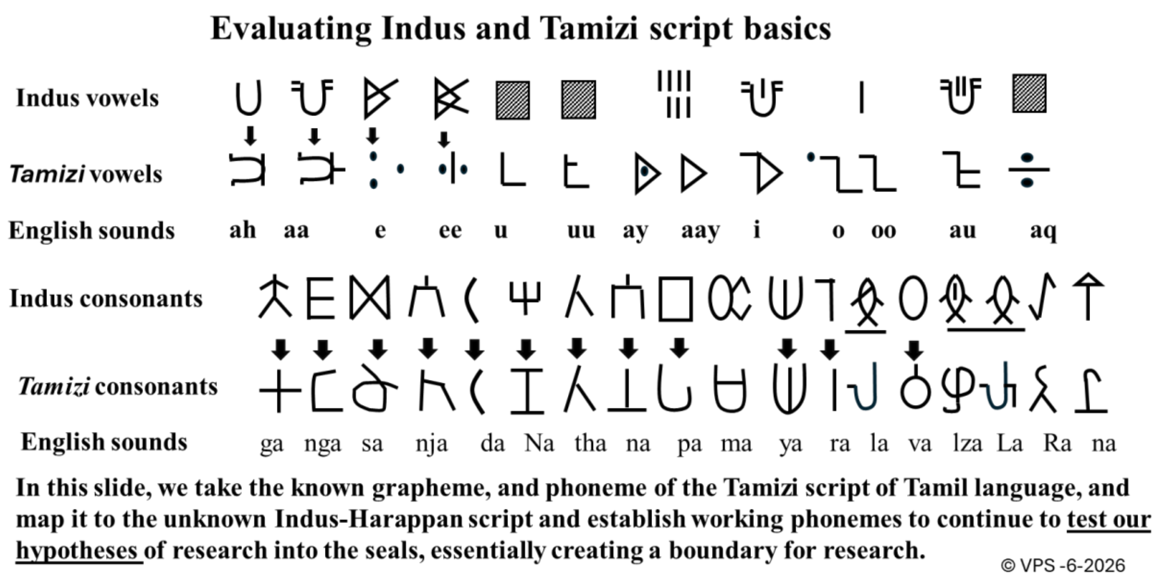


Fig. 2

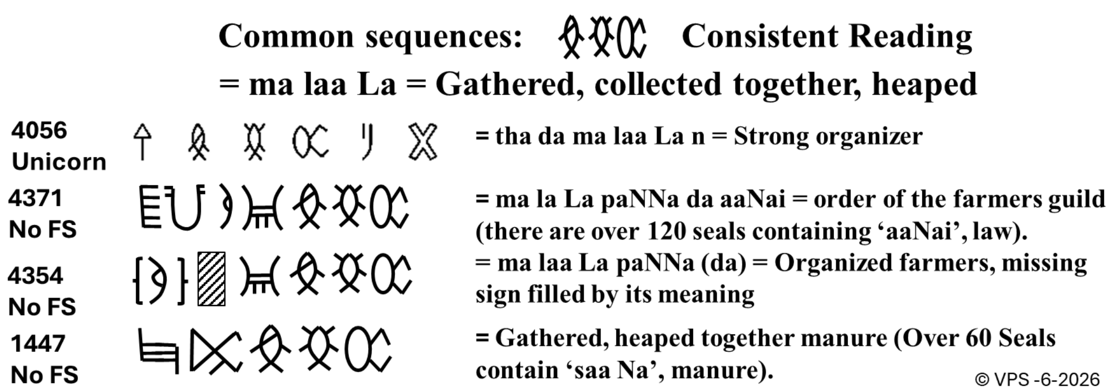
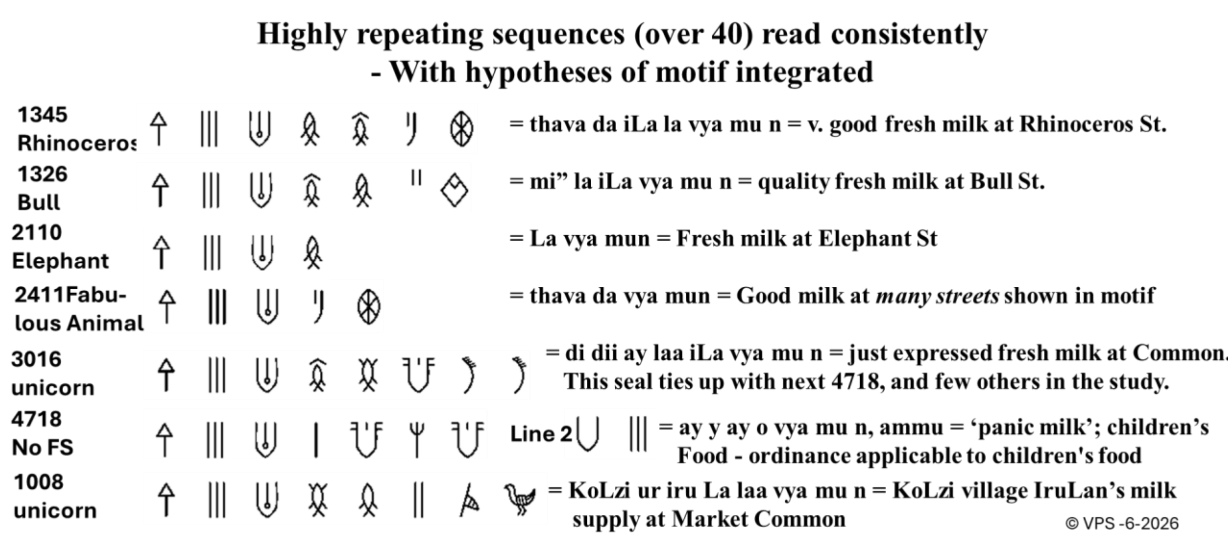


Fig. 3



Figures 4, 5, 6: Script Evolution by Stylization, Acrophony, and Study of Signboard: Fig. 4

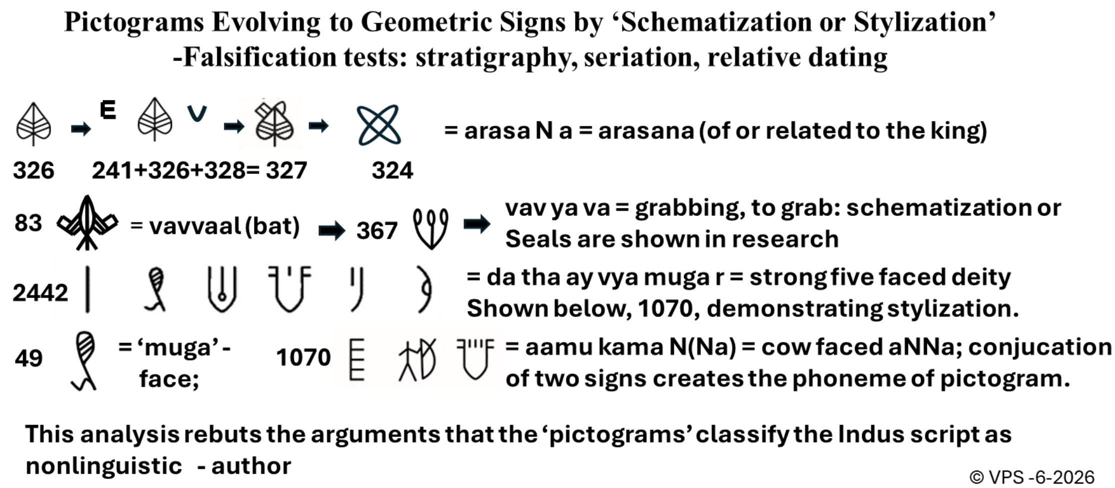


Fig. 5

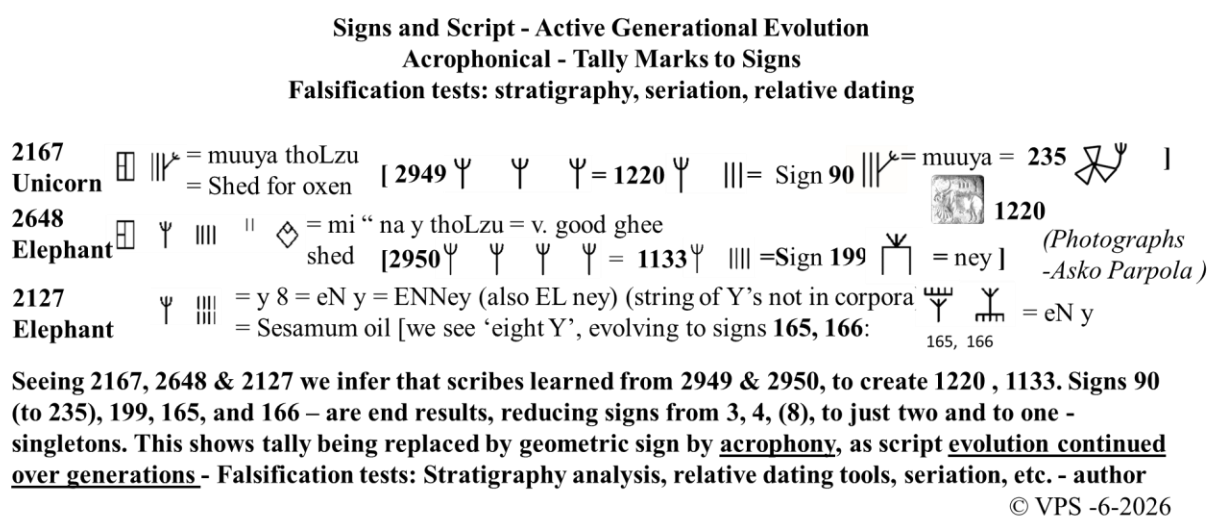


Fig.6

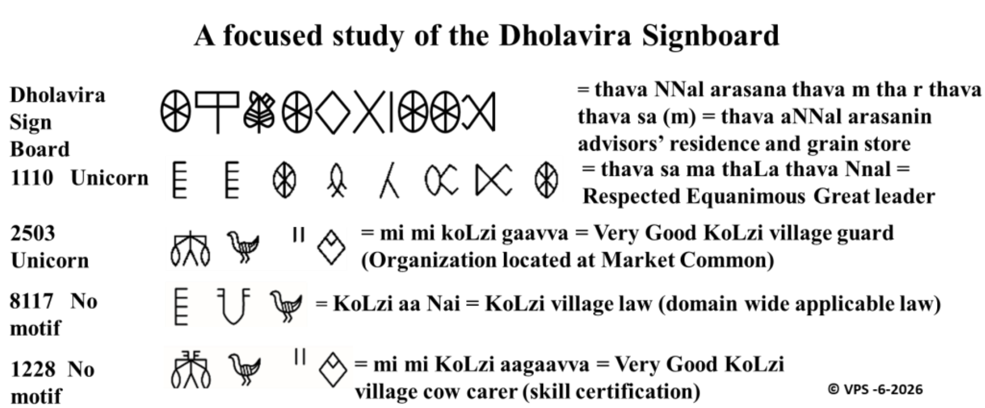
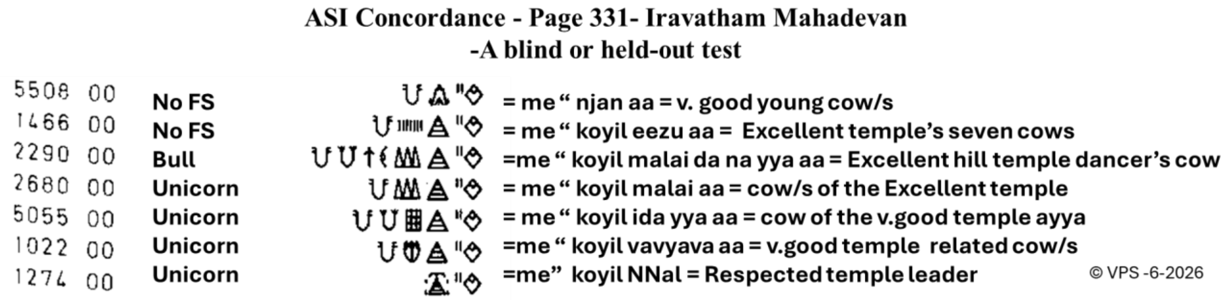


Fig. 7: A sample blind or held-out test: Repeating signs – translated



End Notes

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[i] See note vi

[ii] The jar is the most frequent sign in the Indus Script and hence the discovery of its value must be the first step towards the decipherment of the script, Interpreting the Indus Script: The Dravidian Solution- ***Harappa.com; Iravatham Mahadevan***

[iii] Tholkaappiyam – This ancient Tamil Grammar Treatise is of obscure authorship and age; and all we have are claims and hearsay hypotheses with no discernible proofs other than it is *created* in Tamil and for the Tamil language. This was most probably a grammar *created* (not written) for the spoken language, as speech predates writing by over 200 thousand years.

[iv] A researcher of Indus-Harappan script for over 30 years identified the language aspects of the script and identified Tamil syllabic structure of the script. His research is "sinthuveLiyil munthu thamiz", "*சிந்துவெளியில் முந்து தமிழ்*", Purnachandra Jeeva, Yalisaip Pathippagam, ISBN 978 81 9427 9105, published in 2020 (Tamil)
Current author, though using some of the results, has deviated in many ways, both in phonemes and morphemes in addition to adding considerable depths and opening new grounds like the study of tally marks, modifiers, motifs, and the enlargement of information resulting in increased densities .
Data set 1.0 included, shows the scripts studied and their phonemes, morphemes, in Tamil and English

[v] Most of the Indus-Harappan seals shown in figures are taken from Indus Script Web Tool, Roja Muthiah Research Library (RMRL) Chennai, India (www.indusscript.in)
In figure 3, some seal pictures (e.g. Dholavira sign board) are using the author's Indus Script font designs, used in his publications.
Some seals references without seal numbers but with motifs are taken from,
Corpus of Indus Seals and Inscriptions

(1) Collections in India, edited by Jagat P Pati Joshi and Asko Parpola, Helsinki 1987
(2) Collections in Pakistan, edited by Sayid GG Mustafa Shah and Asko Parpola, Helsinki 1991

[vi] Seals as read and documented in the author's research reports published under different titles:

Ancient Harappans Speak! After 5000 years. ISBN 979-8-9949362-3-5 paperback (Ingram Sparks); 979-8-9940362-4-2, eBook (Amazon)
Indus Script is a Language, A living language of 80 million (Bilingual). ISBN 979-8-9940362-6-6. Self-published 2026, Available at Kavin Publishers: www.kavinpublishing.in

[vii] “The Collapse of the Indus Script Thesis: The myth of a literate Harappan civilization” published jointly by Steve Farmer, Richard Sproat and Michael Witzel in the Electronic Journal of Vedic Studies of 02/11/2004.
[viii] Indus Signs and Graffiti Marks of Tamil Nadu, a morphological Study, K.Rajan, R Sivanantham, Published by Department of archeology, Government of Tamil Nadu, 2025

[ix] Frequency distribution of most occurring signs : Credit Dr. N. Yadav, et all. From Harappa.com

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